

```

PS C:\Users\Chayce\Documents\CollegeFinalSemester\MachineLearning\HW5\HW5\code>
python main.py --task train --run_name my_experiment --data_split simple --
n_layer 2 --n_head 2 --n_embd 16 --max_epochs 60 --batch_size 32 --num_workers 8
--learning_rate 4e-4 --max_len 128 --seed 44 --grad_norm_clip 1.0
Downloading data: 100%|
| 3.27M/3.27M [00:00<00:00, 7.71MB/s]
Downloading data: 100%|
| 812k/812k [00:00<00:00, 7.71MB/s]
Generating train split: 100%|
| 16728/16728 [00:00<00:00, 106008.59 examples/s]
Generating test split: 100%|
| 4182/4182 [00:00<00:00, 89499.14 examples/s]
Building tokenizer at ./tokenizer/simple_vocab.json.
Building tokenizer for actions: 100%|
| 16728/16728 [00:00<00:00, 58382.09it/s]
Building tokenizer for commands: 100%|
| 16728/16728 [00:00<00:00, 60405.24it/s]
tokenizer saved
{'<pad>': 0, '<s>': 1, '</s>': 2, '<unk>': 3, 'I_TURN_RIGHT': 4, 'I_JUMP': 5,
'I_WALK': 6, 'I_TURN_LEFT': 7, 'I_RUN': 8, 'I_LOOK': 9, 'jump': 10, 'opposite':
11, 'right': 12, 'twice': 13, 'and': 14, 'turn': 15, 'thrice': 16, 'run': 17,
'left': 18, 'after': 19, 'walk': 20, 'around': 21, 'look': 22}
train dataset size: 15055
val dataset size: 1673
loading model
total params: 9408
C:
\Users\Chayce\Documents\CollegeFinalSemester\MachineLearning\HW5\HW5\code\traine
r.py:79: FutureWarning: `torch.cuda.amp.GradScaler(args...)` is deprecated.
Please use `torch.amp.GradScaler('cuda', args...)` instead.
    scaler = GradScaler()
C:\Users\Chayce\AppData\Roaming\Python\Python312\site-
packages\torch\amp\grad_scaler.py:132: UserWarning: torch.cuda.amp.GradScaler is
enabled, but CUDA is not available. Disabling.
    warnings.warn(
epoch 1 iter 0: train loss 0.000000. lr 0.0000e+00: 0%|
| 0/471 [00:00<?, ?it/s]C
:
\Users\Chayce\Documents\CollegeFinalSemester\MachineLearning\HW5\HW5\code\traine
r.py:102: FutureWarning: `torch.cuda.amp.autocast(args...)` is deprecated.
Please use `torch.amp.autocast('cuda', args...)` instead.
    with torch.cuda.amp.autocast():
C:\Users\Chayce\AppData\Roaming\Python\Python312\site-
packages\torch\amp\autocast_mode.py:266: UserWarning: User provided device_type
of 'cuda', but CUDA is not available. Disabling
    warnings.warn(
epoch 1 iter 470: train loss 1.35032. lr 3.9978e-04: 100%|
| 471/471 [00:19<00:00, 24.66it/s]
test loss: %f 1.325701360432607
epoch_valid_loss: 1.325701360432607, epoch_train_loss: 2.00851627747724, epoch:
1
Saving at epoch 1:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 1.3370606899261475 train_step: 500, learning_rate:
0.0003997489407867487
27/471 [00:01<00:16, 26.59it/s]

```

epoch 2 iter 470: train loss 0.84570. lr 3.9902e-04: 100%|

| 471/471 [00:18<00:00, 25.69it/s]

test loss: %f 0.8321844598032394

epoch_valid_loss: 0.8321844598032394, epoch_train_loss: 1.0785905755502656,
epoch: 2

Saving at epoch 2:

./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt

step_train_loss: 0.9210577011108398 train_step: 1000, learning_rate:
0.0003988910329278014

57/471 [00:02<00:18, 22.13it/s]

epoch 3 iter 470: train loss 0.53342. lr 3.9773e-04: 100%|

| 471/471 [00:20<00:00, 22.63it/s]

test loss: %f 0.6822825321611369

epoch_valid_loss: 0.6822825321611369, epoch_train_loss: 0.8280024045085704,
epoch: 3

Saving at epoch 3:

./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt

step_train_loss: 0.792720377445221 train_step: 1500, learning_rate:
0.00039742625820294794

87/471 [00:03<00:14, 26.68it/s]

epoch 4 iter 470: train loss 0.84916. lr 3.9590e-04: 100%|

| 471/471 [00:18<00:00, 25.07it/s]

test loss: %f 0.6435250504961554

epoch_valid_loss: 0.6435250504961554, epoch_train_loss: 0.7493402876924811,
epoch: 4

Saving at epoch 4:

./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt

step_train_loss: 0.7068024277687073 train_step: 2000, learning_rate:
0.00039535908601049877

114/471 [00:04<00:13, 25.90it/s]

epoch 5 iter 470: train loss 0.86418. lr 3.9354e-04: 100%|

| 471/471 [00:19<00:00, 23.93it/s]

test loss: %f 0.6127927983706852

epoch_valid_loss: 0.6127927983706852, epoch_train_loss: 0.7094877634331932,
epoch: 5

Saving at epoch 5:

./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt

step_train_loss: 0.6610397696495056 train_step: 2500, learning_rate:
0.0003926958238158596

144/471 [00:05<00:12, 27.04it/s]

epoch 6 iter 470: train loss 0.64282. lr 3.9065e-04: 100%|

| 471/471 [00:19<00:00, 24.72it/s]

test loss: %f 0.587410347641639

epoch_valid_loss: 0.587410347641639, epoch_train_loss: 0.6883237287243966,
epoch: 6

Saving at epoch 6:

./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt

step_train_loss: 0.5538056492805481 train_step: 3000, learning_rate:
0.00038944459790585885

174/471 [00:06<00:11, 26.38it/s]

epoch 7 iter 470: train loss 0.55695. lr 3.8725e-04: 100%|

| 471/471 [00:18<00:00, 25.07it/s]

test loss: %f 0.5606776009190757

epoch_valid_loss: 0.5606776009190757, epoch_train_loss: 0.666973089716237,
epoch: 7

Saving at epoch 7:

./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt

step_train_loss: 0.6967178583145142 train_step: 3500, learning_rate:

```
0.00038561532859338987
201/471 [00:07<00:10, 26.57it/s]
epoch 8 iter 470: train loss 0.81323. lr 3.8334e-04: 100%|
| 471/471 [00:18<00:00, 25.36it/s]
test loss: %f 0.5387572258148553
epoch_valid_loss: 0.5387572258148553, epoch_train_loss: 0.6498176260612066,
epoch: 8
Saving at epoch 8:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6844774484634399 train_step: 4000, learning_rate:
0.00038121969994802686
231/471 [00:09<00:08, 26.73it/s]
epoch 9 iter 470: train loss 0.50556. lr 3.7894e-04: 100%|
| 471/471 [00:19<00:00, 24.54it/s]
test loss: %f 0.5172963811541503
epoch_valid_loss: 0.5172963811541503, epoch_train_loss: 0.6367730285078603,
epoch: 9
Saving at epoch 9:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6613866090774536 train_step: 4500, learning_rate:
0.00037627112671667753
261/471 [00:10<00:08, 24.64it/s]
epoch 10 iter 470: train loss 0.62629. lr 3.7405e-04: 100%|
| 471/471 [00:18<00:00, 24.84it/s]
test loss: %f 0.4982761453907445
epoch_valid_loss: 0.4982761453907445, epoch_train_loss: 0.6244264639233834,
epoch: 10
Saving at epoch 10:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5298473834991455 train_step: 5000, learning_rate:
0.0003707847005411132
288/471 [00:11<00:06, 26.50it/s]
epoch 11 iter 470: train loss 0.56648. lr 3.6869e-04: 100%|
| 471/471 [00:18<00:00, 24.82it/s]
test loss: %f 0.4868178918676556
epoch_valid_loss: 0.4868178918676556, epoch_train_loss: 0.613663550268566,
epoch: 11
Saving at epoch 11:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5498161911964417 train_step: 5500, learning_rate:
0.0003647771665180489
318/471 [00:12<00:05, 26.57it/s]
epoch 12 iter 470: train loss 0.66675. lr 3.6287e-04: 100%|
| 471/471 [00:19<00:00, 24.23it/s]
test loss: %f 0.4866598462158779
epoch_valid_loss: 0.4866598462158779, epoch_train_loss: 0.6062141630933037,
epoch: 12
Saving at epoch 12:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5661967992782593 train_step: 6000, learning_rate:
0.0003582668618277934
348/471 [00:13<00:04, 26.81it/s]
epoch 13 iter 470: train loss 0.66951. lr 3.5661e-04: 100%|
| 471/471 [00:19<00:00, 24.72it/s]
test loss: %f 0.4805772456358064
epoch_valid_loss: 0.4805772456358064, epoch_train_loss: 0.5987639141816749,
epoch: 13
Saving at epoch 13:
```

```
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5967237949371338 train_step: 6500, learning_rate:
0.00035127364537228863
375/471 [00:14<00:03, 26.94it/s]
epoch 14 iter 470: train loss 0.71236. lr 3.4993e-04: 100%|
| 471/471 [00:18<00:00, 25.33it/s]
test loss: %f 0.4636498897705438
epoch_valid_loss: 0.4636498897705438, epoch_train_loss: 0.5908579540353925,
epoch: 14
Saving at epoch 14:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.574994683265686 train_step: 7000, learning_rate:
0.00034381884763814557
405/471 [00:15<00:02, 26.53it/s]
epoch 15 iter 470: train loss 0.56564. lr 3.4284e-04: 100%|
| 471/471 [00:18<00:00, 25.42it/s]
test loss: %f 0.4631116750105372
epoch_valid_loss: 0.4631116750105372, epoch_train_loss: 0.5868301976258588,
epoch: 15
Saving at epoch 15:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6047993302345276 train_step: 7500, learning_rate:
0.00033592522177779913
435/471 [00:16<00:01, 26.98it/s]
epoch 16 iter 470: train loss 0.51144. lr 3.3536e-04: 100%|
| 471/471 [00:18<00:00, 25.29it/s]
test loss: %f 0.44472328671869243
epoch_valid_loss: 0.44472328671869243, epoch_train_loss: 0.584938205857692,
epoch: 16
Saving at epoch 16:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5582631230354309 train_step: 8000, learning_rate:
0.0003276168616470794
462/471 [00:17<00:00, 26.99it/s]
epoch 17 iter 470: train loss 0.52281. lr 3.2752e-04: 100%|
| 471/471 [00:18<00:00, 25.37it/s]
test loss: %f 0.45056670173159186
epoch_valid_loss: 0.45056670173159186, epoch_train_loss: 0.5784837742773085,
epoch: 17
epoch 18 iter 470: train loss 0.51078. lr 3.1934e-04: 100%|
| 471/471 [00:18<00:00, 25.24it/s]
test loss: %f 0.4397388156854881
epoch_valid_loss: 0.4397388156854881, epoch_train_loss: 0.5733783177382881,
epoch: 18
Saving at epoch 18:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5079430341720581 train_step: 8500, learning_rate:
0.00031892856370618823
21/471 [00:00<00:17, 26.41it/s]
epoch 19 iter 470: train loss 0.45407. lr 3.1083e-04: 100%|
| 471/471 [00:18<00:00, 25.19it/s]
test loss: %f 0.4421397087708959
epoch_valid_loss: 0.4421397087708959, epoch_train_loss: 0.5669960694707883,
epoch: 19
step_train_loss: 0.48146703839302063 train_step: 9000, learning_rate:
0.00030986831410617134
51/471 [00:01<00:15, 26.98it/s]
epoch 20 iter 470: train loss 0.60223. lr 3.0202e-04: 100%|
```

```
| 471/471 [00:18<00:00, 25.41it/s]
test loss: %f 0.42118817230440536
epoch_valid_loss: 0.42118817230440536, epoch_train_loss: 0.5611753949693813,
epoch: 20
Saving at epoch 20:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6194795966148376 train_step: 9500, learning_rate:
0.0003004728463160256
78/471 [00:02<00:14, 26.84it/s]
epoch 21 iter 470: train loss 0.53945. lr 2.9293e-04: 100%|
```

```
| 471/471 [00:18<00:00, 25.33it/s]
test loss: %f 0.4049304763101182
epoch_valid_loss: 0.4049304763101182, epoch_train_loss: 0.5549289967603744,
epoch: 21
Saving at epoch 21:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5636781454086304 train_step: 10000, learning_rate:
0.00029077081075159177
108/471 [00:04<00:15, 23.79it/s]
epoch 22 iter 470: train loss 0.60799. lr 2.8359e-04: 100%|
```

```
| 471/471 [00:19<00:00, 23.95it/s]
test loss: %f 0.39651060497985696
epoch_valid_loss: 0.39651060497985696, epoch_train_loss: 0.5510887294185136,
epoch: 22
Saving at epoch 22:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5360615253448486 train_step: 10500, learning_rate:
0.00028079179083144917
138/471 [00:05<00:12, 26.27it/s]
epoch 23 iter 470: train loss 0.65390. lr 2.7403e-04: 100%|
```

```
| 471/471 [00:18<00:00, 25.19it/s]
test loss: %f 0.39165080155966414
epoch_valid_loss: 0.39165080155966414, epoch_train_loss: 0.541974545917187,
epoch: 23
Saving at epoch 23:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6054497957229614 train_step: 11000, learning_rate:
0.00027056627435953035
165/471 [00:06<00:12, 25.45it/s]
epoch 24 iter 470: train loss 0.46554. lr 2.6427e-04: 100%|
```

```
| 471/471 [00:19<00:00, 24.77it/s]
test loss: %f 0.3865899870980461
epoch_valid_loss: 0.3865899870980461, epoch_train_loss: 0.5358795519214273,
epoch: 24
Saving at epoch 24:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5403819680213928 train_step: 11500, learning_rate:
0.00026012544254159104
195/471 [00:07<00:10, 26.25it/s]
epoch 25 iter 470: train loss 0.57616. lr 2.5433e-04: 100%|
```

```
| 471/471 [00:18<00:00, 25.20it/s]
test loss: %f 0.3753447251499824
epoch_valid_loss: 0.3753447251499824, epoch_train_loss: 0.5288762243943103,
epoch: 25
Saving at epoch 25:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.4979563355445862 train_step: 12000, learning_rate:
0.00024950115296592005
```

```
225/471 [00:08<00:09, 25.60it/s]
epoch 26 iter 470: train loss 0.52188. lr 2.4424e-04: 100%|
██████████| 471/471 [00:19<00:00, 24.45it/s]
test loss: %f 0.371193398844521
epoch_valid_loss: 0.371193398844521, epoch_train_loss: 0.5185199732993059,
epoch: 26
Saving at epoch 26:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5346850156784058 train_step: 12500, learning_rate:
0.00023872582299629312██████████
252/471 [00:09<00:08, 26.07it/s]
epoch 27 iter 470: train loss 0.65860. lr 2.3404e-04: 100%|
██████████| 471/471 [00:19<00:00, 24.55it/s]
test loss: %f 0.3511817455291748
epoch_valid_loss: 0.3511817455291748, epoch_train_loss: 0.5089660602010739,
epoch: 27
Saving at epoch 27:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.6776638627052307 train_step: 13000, learning_rate:
0.0002278323092971486██████████
282/471 [00:10<00:07, 26.51it/s]
epoch 28 iter 470: train loss 0.55112. lr 2.2374e-04: 100%|
██████████| 471/471 [00:18<00:00, 25.04it/s]
test loss: %f 0.34643837753331885
epoch_valid_loss: 0.34643837753331885, epoch_train_loss: 0.5061572997686463,
epoch: 28
Saving at epoch 28:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.5411255955696106 train_step: 13500, learning_rate:
0.00021685389362485693██████████
312/471 [00:12<00:07, 20.23it/s]
epoch 29 iter 470: train loss 0.44995. lr 2.1338e-04: 100%|
██████████| 471/471 [00:20<00:00, 22.73it/s]
test loss: %f 0.3361074227207112
epoch_valid_loss: 0.3361074227207112, epoch_train_loss: 0.49842881352785035,
epoch: 29
Saving at epoch 29:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.4873189628124237 train_step: 14000, learning_rate:
0.00020582405250931256██████████
339/471 [00:13<00:05, 25.56it/s]
epoch 30 iter 470: train loss 0.42239. lr 2.0298e-04: 100%|
██████████| 471/471 [00:19<00:00, 24.20it/s]
test loss: %f 0.3235594380014348
epoch_valid_loss: 0.3235594380014348, epoch_train_loss: 0.492399578008429,
epoch: 30
Saving at epoch 30:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.4796505570411682 train_step: 14500, learning_rate:
0.0001947764189876604██████████
369/471 [00:14<00:03, 25.55it/s]
epoch 31 iter 470: train loss 0.46406. lr 1.9258e-04: 100%|
██████████| 471/471 [00:19<00:00, 24.23it/s]
test loss: %f 0.3274189200041429
epoch_valid_loss: 0.3274189200041429, epoch_train_loss: 0.4850165375099061,
epoch: 31
step_train_loss: 0.5144190788269043 train_step: 15000, learning_rate:
0.00018374474568164296██████████
```

```
399/471 [00:15<00:02, 25.59it/s]
epoch 32 iter 470: train loss 0.59784. lr 1.8219e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 471/471 [00:19<00:00, 24.27it/s]
test loss: %f 0.3281871024167763
epoch_valid_loss: 0.3281871024167763, epoch_train_loss: 0.4815313314303218,
epoch: 32
step_train_loss: 0.4622149169445038 train_step: 15500, learning_rate:
0.00017276267122068255████████████████████████████████████████████████████████████████████████████████
426/471 [00:16<00:01, 25.15it/s]
epoch 33 iter 470: train loss 0.42052. lr 1.7185e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 471/471 [00:19<00:00, 24.02it/s]
test loss: %f 0.3106205244109316
epoch_valid_loss: 0.3106205244109316, epoch_train_loss: 0.472520410698184,
epoch: 33
Saving at epoch 33:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.4669337570667267 train_step: 16000, learning_rate:
0.000161863704659693████████████████████████████████████████████████████████████████████████████████
456/471 [00:17<00:00, 26.51it/s]
epoch 34 iter 470: train loss 0.47234. lr 1.6159e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 471/471 [00:18<00:00, 25.19it/s]
test loss: %f 0.30425603659647815
epoch_valid_loss: 0.30425603659647815, epoch_train_loss: 0.47220033774203807,
epoch: 34
Saving at epoch 34:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
epoch 35 iter 470: train loss 0.54613. lr 1.5144e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 471/471 [00:18<00:00, 25.11it/s]
test loss: %f 0.2938822311050487
epoch_valid_loss: 0.2938822311050487, epoch_train_loss: 0.4659147669428726,
epoch: 35
Saving at epoch 35:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.4304744601249695 train_step: 16500, learning_rate:
0.0001510924808426786████████████████████████████████████████████████████████████████████████████████
15/471 [00:00<00:17, 26.58it/s]
epoch 36 iter 470: train loss 0.40472. lr 1.4141e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 471/471 [00:18<00:00, 25.24it/s]
test loss: %f 0.2938516637626684
epoch_valid_loss: 0.2938516637626684, epoch_train_loss: 0.46155175560345824,
epoch: 36
Saving at epoch 36:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.49568697810173035 train_step: 17000, learning_rate:
0.00014045898638681664████████████████████████████████████████████████████████████████████████████████
42/471 [00:01<00:16, 25.38it/s]
epoch 37 iter 470: train loss 0.40144. lr 1.3154e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 471/471 [00:18<00:00, 25.80it/s]
test loss: %f 0.2955200177318645
epoch_valid_loss: 0.2955200177318645, epoch_train_loss: 0.45813025632228566,
epoch: 37
step_train_loss: 0.4173870086669922 train_step: 17500, learning_rate:
0.00013000710454508208████████████████████████████████████████████████████████████████████████████████
72/471 [00:02<00:15, 26.50it/s]
epoch 38 iter 470: train loss 0.31777. lr 1.2186e-04: 100%|
████████████████████████████████████████████████████████████████████████████████
██████████ | 471/471 [00:18<00:00, 25.12it/s]
test loss: %f 0.28481422167903975
```

```
epoch_valid_loss: 0.28481422167903975, epoch_train_loss: 0.45177345873190844,  
epoch: 38  
Saving at epoch 38:  
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt  
step_train_loss: 0.47447478771209717 train_step: 18000, learning_rate:  
0.00011976883009889161  
100/471 [00:04<00:13, 26.69it/s]  
epoch 39 iter 470: train loss 0.36266. lr 1.1239e-04: 100%|  
████████████████████████████████████████████████████████████████████████████████  
██████████| 471/471 [00:19<00:00, 24.41it/s]  
test loss: %f 0.2908231258954642  
epoch_valid_loss: 0.2908231258954642, epoch_train_loss: 0.4498369581760115,  
epoch: 39  
step_train_loss: 0.4082123935222626 train_step: 18500, learning_rate:  
0.00010977536101143186  
129/471 [00:04<00:12, 27.86it/s]  
epoch 40 iter 470: train loss 0.51640. lr 1.0316e-04: 100%|  
████████████████████████████████████████████████████████████████████████████████  
██████████| 471/471 [00:17<00:00, 26.32it/s]  
test loss: %f 0.27767015599979544  
epoch_valid_loss: 0.27767015599979544, epoch_train_loss: 0.4475993465972301,  
epoch: 40  
Saving at epoch 40:  
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt  
step_train_loss: 0.4815995991230011 train_step: 19000, learning_rate:  
0.00010005718985545179  
159/471 [00:06<00:12, 25.34it/s]  
epoch 41 iter 470: train loss 0.53378. lr 9.4187e-05: 100%|  
████████████████████████████████████████████████████████████████████████████████  
██████████| 471/471 [00:18<00:00, 25.00it/s]  
test loss: %f 0.27365161394173243  
epoch_valid_loss: 0.27365161394173243, epoch_train_loss: 0.4441494408306802,  
epoch: 41  
Saving at epoch 41:  
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt  
step_train_loss: 0.4456326961517334 train_step: 19500, learning_rate:  
9.064396920088103e-05  
189/471 [00:07<00:10, 26.68it/s]  
epoch 42 iter 470: train loss 0.46040. lr 8.5503e-05: 100%|  
████████████████████████████████████████████████████████████████████████████████  
██████████| 471/471 [00:18<00:00, 25.17it/s]  
test loss: %f 0.2748026091535136  
epoch_valid_loss: 0.2748026091535136, epoch_train_loss: 0.44281556050742, epoch:  
42  
step_train_loss: 0.4142102599143982 train_step: 20000, learning_rate:  
8.156442113742624e-05  
216/471 [00:08<00:09, 27.10it/s]  
epoch 43 iter 470: train loss 0.38690. lr 7.7128e-05: 100%|  
████████████████████████████████████████████████████████████████████████████████  
██████████| 471/471 [00:18<00:00, 25.18it/s]  
test loss: %f 0.2705690658317422  
epoch_valid_loss: 0.2705690658317422, epoch_train_loss: 0.43776908839584155,  
epoch: 43  
Saving at epoch 43:  
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt  
step_train_loss: 0.4504488706588745 train_step: 20500, learning_rate:  
7.284624963629354e-05  
246/471 [00:09<00:08, 26.37it/s]  
epoch 44 iter 470: train loss 0.45526. lr 6.9086e-05: 100%|  
████████████████████████████████████████████████████████████████████████████████  
██████████| 471/471 [00:18<00:00, 25.24it/s]  
test loss: %f 0.2741787413943489  
epoch_valid_loss: 0.2741787413943489, epoch_train_loss: 0.4389552083617816,  
epoch: 44
```

```
step_train_loss: 0.3776760697364807 train_step: 21000, learning_rate:
6.451602398585763e-05 [REDACTED] |
275/471 [00:09<00:06, 28.23it/s]
epoch 45 iter 470: train loss 0.47805. lr 6.1399e-05: 100%|
[REDACTED] | 471/471 [00:17<00:00, 26.41it/s]
test loss: %f 0.2706576636377371
epoch_valid_loss: 0.2706576636377371, epoch_train_loss: 0.4345495940014056,
epoch: 45
step_train_loss: 0.4129829704761505 train_step: 21500, learning_rate:
5.659922743246404e-05 [REDACTED] |
303/471 [00:10<00:06, 27.97it/s]
epoch 46 iter 470: train loss 0.36016. lr 5.4087e-05: 100%|
[REDACTED] | 471/471 [00:17<00:00, 26.50it/s]
test loss: %f 0.2663930873825865
epoch_valid_loss: 0.2663930873825865, epoch_train_loss: 0.4341250416959167,
epoch: 46
Saving at epoch 46:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.3631543219089508 train_step: 22000, learning_rate:
4.911998248917342e-05 [REDACTED] |
333/471 [00:11<00:04, 28.09it/s]
epoch 47 iter 470: train loss 0.35812. lr 4.7170e-05: 100%|
[REDACTED] | 471/471 [00:17<00:00, 26.30it/s]
test loss: %f 0.26772191670705686
epoch_valid_loss: 0.26772191670705686, epoch_train_loss: 0.4340799409745858,
epoch: 47
step_train_loss: 0.4558402895927429 train_step: 22500, learning_rate:
4.210111020226544e-05 [REDACTED] |
363/471 [00:13<00:03, 28.36it/s]
epoch 48 iter 470: train loss 0.59606. lr 4.0667e-05: 100%|
[REDACTED] | 471/471 [00:17<00:00, 26.37it/s]
test loss: %f 0.26648200739104794
epoch_valid_loss: 0.26648200739104794, epoch_train_loss: 0.4338262744896478,
epoch: 48
step_train_loss: 0.44047266244888306 train_step: 23000, learning_rate:
4e-05 [REDACTED] |
392/471 [00:14<00:02, 28.96it/s]
epoch 49 iter 470: train loss 0.49366. lr 4.0000e-05: 100%|
[REDACTED] | 471/471 [00:17<00:00, 26.46it/s]
test loss: %f 0.2622922197827753
epoch_valid_loss: 0.2622922197827753, epoch_train_loss: 0.43061657012647886,
epoch: 49
Saving at epoch 49:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.41518139839172363 train_step: 23500, learning_rate:
4e-05 [REDACTED] |
419/471 [00:15<00:01, 28.05it/s]
epoch 50 iter 470: train loss 0.36489. lr 4.0000e-05: 100%|
[REDACTED] | 471/471 [00:17<00:00, 26.46it/s]
test loss: %f 0.2674879378305291
epoch_valid_loss: 0.2674879378305291, epoch_train_loss: 0.4300348288567456,
epoch: 50
step_train_loss: 0.48105305433273315 train_step: 24000, learning_rate:
4e-05 [REDACTED] |
450/471 [00:16<00:00, 26.79it/s]
epoch 51 iter 470: train loss 0.44192. lr 4.0000e-05: 100%|
[REDACTED] | 471/471 [00:17<00:00, 26.46it/s]
```

test loss: %f 0.26266990686362646
epoch_valid_loss: 0.26266990686362646, epoch_train_loss: 0.43014180280600384,
epoch: 51
epoch 52 iter 470: train loss 0.36469. lr 4.0000e-05: 100%|

| 471/471 [00:17<00:00, 26.81it/s]
test loss: %f 0.2634043207146087
epoch_valid_loss: 0.2634043207146087, epoch_train_loss: 0.4303104631333847,
epoch: 52
step_train_loss: 0.46717730164527893 train_step: 24500, learning_rate: 4e-05
| 6/471 [00:00<00:16, 28.04it/s]
epoch 53 iter 470: train loss 0.57184. lr 4.0000e-05: 100%|

| 471/471 [00:17<00:00, 26.87it/s]
test loss: %f 0.260306271460821
epoch_valid_loss: 0.260306271460821, epoch_train_loss: 0.4289506557387658,
epoch: 53
Saving at epoch 53:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.3948806822299957 train_step: 25000, learning_rate: 4e-05
| 36/471 [00:01<00:15, 28.35it/s]
epoch 54 iter 470: train loss 0.32274. lr 4.0000e-05: 100%|

| 471/471 [00:17<00:00, 26.83it/s]
test loss: %f 0.2640353787057805
epoch_valid_loss: 0.2640353787057805, epoch_train_loss: 0.4262873923955703,
epoch: 54
step_train_loss: 0.3913167715072632 train_step: 25500, learning_rate: 4e-05
| 64/471 [00:02<00:14, 27.82it/s]
epoch 55 iter 470: train loss 0.38310. lr 4.0000e-05: 100%|

| 471/471 [00:17<00:00, 26.96it/s]
test loss: %f 0.26188740190469995
epoch_valid_loss: 0.26188740190469995, epoch_train_loss: 0.4284657799261138,
epoch: 55
step_train_loss: 0.42548754811286926 train_step: 26000, learning_rate: 4e-05
| 93/471 [00:03<00:13, 28.41it/s]
epoch 56 iter 470: train loss 0.34689. lr 4.0000e-05: 100%|

| 471/471 [00:17<00:00, 26.61it/s]
test loss: %f 0.2577916184123957
epoch_valid_loss: 0.2577916184123957, epoch_train_loss: 0.426970814380423,
epoch: 56
Saving at epoch 56:
./cond_gpt/weights/my_experiment_simplesplit_2layer_2head_16embd_32bs.pt
step_train_loss: 0.36751508712768555 train_step: 26500, learning_rate:
4e-05
124/471 [00:04<00:12, 28.58it/s]
epoch 57 iter 470: train loss 0.44393. lr 4.0000e-05: 100%|

| 471/471 [00:17<00:00, 26.81it/s]
test loss: %f 0.2621100890748906
epoch_valid_loss: 0.2621100890748906, epoch_train_loss: 0.4276491145419467,
epoch: 57
step_train_loss: 0.47926121950149536 train_step: 27000, learning_rate:
4e-05
151/471 [00:05<00:11, 28.09it/s]
epoch 58 iter 470: train loss 0.39307. lr 4.0000e-05: 100%|

| 471/471 [00:17<00:00, 26.63it/s]
test loss: %f 0.26155544255139695
epoch_valid_loss: 0.26155544255139695, epoch_train_loss: 0.4258403503464539,
epoch: 58
step_train_loss: 0.39314109086990356 train_step: 27500, learning_rate:

4e-05 |
180/471 [00:06<00:10, 28.72it/s]
epoch 59 iter 470: train loss 0.36448. lr 4.0000e-05: 100%|
| 471/471 [00:17<00:00, 26.72it/s]
test loss: %f 0.26124161397511103
epoch_valid_loss: 0.26124161397511103, epoch_train_loss: 0.4227984719721881,
epoch: 59
step_train_loss: 0.4268701374530792 train_step: 28000, learning_rate:
4e-05 |
211/471 [00:07<00:09, 28.20it/s]
epoch 60 iter 470: train loss 0.41615. lr 4.0000e-05: 100%|
| 471/471 [00:17<00:00, 26.86it/s]
test loss: %f 0.2590955774176796
epoch_valid_loss: 0.2590955774176796, epoch_train_loss: 0.42673711545148474,
epoch: 60